RIDHANYA S

MACHINE LEARNING ENGINEER

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ridhanya-s

TECHNICAL SKILLS :

- Languages: Python, JavaScript (Basics)
- LLMs & GenAI: GPT, Claude, Gemini, LLaMA, Mistral, Hugging Face Models
- ML Frameworks & Libraries: Keras, TensorFlow, Scikit-learn, NumPy, Pandas
- Agent Frameworks: LangChain, LangGraph, Multi-Agent Orchestration
- NLP & Speech: NLTK, spaCy, OpenAI Whisper, NLP Fundamentals
- Vector Databases: Pinecone, ChromaDB
- Model Evaluation: BLEU, ROUGE, METEOR, BERTScore, A/B Testing, Prompt Engineering
- APIs & Backend: FastAPI, Flask, Streamlit, Pydantic, WebSockets
- Cloud & Services: AWS (S3, DynamoDB, Textract), Azure (OpenAI, Computer Vision, Cognitive Services, Blob Storage, Azure Function)
- Databases: MySQL, PostgreSQL, DynamoDB, MongoDB
- Testing & Code Quality: Pytest, SonarQube, Ruff
- Other Tools: GitHub, VS Code, Jupyter Notebook, OpenCV
- Domain Knowledge :Generative AI, Computer Vision, NLP, Machine Learning, Deep Learning

PROFILE:

Machine Learning Engineer with 2.5+ years of experience designing and delivering end-to-end AI systems across Generative AI, LLM fine-tuning, Multi-Agent architectures, RAG pipelines, NLP, and speech-to-text integration. Proficient in Python-based AI/ML pipeline development using LangChain, LangGraph, PyTorch, and Hugging Face Transformers, with hands-on experience integrating AWS and Azure services. Skilled in working with leading LLMs including GPT, Claude, and Gemini, as well as open-source models such as LLaMA, Mistral. Brings a structured, experiment-driven approach to model evaluation, prompt engineering, and continuous improvement with a strong track record of translating complex AI capabilities into reliable, production-grade solutions.

EXPERIENCE:

MACHINE LEARNING ENGINEER I 2024 - PRESENT

OptiSol Business Solutions Pvt. Ltd.

- Recognised with 2 Spot Awards for technical excellence and consistent project delivery.
- Architected an end-to-end AI pipeline with audio capture, Whisper STT, LLM summarisation, and structured report generation integrated with Azure Cognitive Services.
- Designed stateful Multi-Agent systems using LangGraph, enabling controlled LLM reasoning and context-aware enterprise workflow processing.
- Fine-tuned open-source LLMs (LLaMA, Mistral, T5, LSTM) for domain-specific use cases; performance evaluation using BLEU, ROUGE, and BERTScore.
- Built OCR and Vision-LLM pipelines for document and seal verification using Azure Computer Vision and AWS Textract, improving accuracy for low-quality inputs.
- Engineered prompt strategies, evaluation frameworks, and Pydantic-based schema validation to ensure structured, token-efficient model outputs.
- Built and maintained production-grade REST APIs using FastAPI, integrating AWS and Azure services with asynchronous processing and WebSocket support.
- Automated meeting-insight extraction and task lifecycle management by integrating Microsoft Graph API and third-party platforms within LangChain agentic workflows.
- Upheld engineering standards through CI/CD maintenance, Pytest-based test coverage, code quality reviews (SonarQube, Ruff), and structured logging for observability.

E D U C A T I O N :

M.SC. ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Coimbatore Institute of Technology (CIT), Coimbatore

P R O J E C T S :

INTELLIGENT MEETING INSIGHTS & ACTION MANAGEMENT SYSTEM

Python · FastAPI · OpenAI Whisper · Azure OpenAI · Microsoft Graph API · Trello API · Confluence API

- Developed an automated system to extract and manage meeting insights from audio recordings and transcripts using OpenAI Whisper for Speech-to-Text transcription.
- Implemented a multi-layer validation pipeline (LLM, regex, schema, business rules) to extract structured outputs with action items, owners, and deadlines.
- Integrated Trello and Confluence APIs for real-time task creation and lifecycle management.
- Designed a provider-agnostic LLM architecture supporting OpenAI, Azure OpenAI, and open-source models with asynchronous backend processing.

DOCUMENT INTELLIGENCE & VALIDATION SYSTEM

Python · FastAPI · LangGraph · LangChain · Azure OpenAI · AWS Textract · DynamoDB

- Architected an AI-driven platform to process and validate large-scale enterprise documents using a stateful Multi-Agent workflow with LangGraph.
- Integrated hybrid OCR and LLM pipelines (AWS Textract + Azure OpenAI) to enhance semantic understanding beyond raw text extraction.
- Enforced structured and deterministic outputs using Pydantic schema-constrained prompts with confidence-based post-processing for multi-page inconsistencies.
- Incorporated PII masking, audit-ready state persistence, observability, and retry mechanisms for production-grade reliability.

LEGAL DISPUTE ANALYZER

Python · FastAPI · LangChain · Azure OpenAI · Google Gemini · ChromaDB · AWS S3 · ReportLab

- Built an AI-powered legal document analysis system with a hybrid RAG pipeline combining BM25 and dense embeddings for high-accuracy retrieval.
- Designed a multi-LLM architecture with dynamic switching between Azure OpenAI and Gemini based on task requirements and cost efficiency.
- Exposed functionalities via FastAPI endpoints with structured error handling and retry mechanisms.
- Generated professional PDF reports with citation handling, chronology extraction, and structured summaries, significantly reducing manual legal review effort.

AI-BASED COI PROCESSING SYSTEM

React · FastAPI · PostgreSQL · Azure OpenAI · Azure Computer Vision · WebSockets · LLM Integration

- Developed a full-stack application to automate Certificate of Insurance workflows with a React frontend and FastAPI backend.
- Built REST APIs with authentication, structured request/response handling, and real-time updates via WebSockets, backed by PostgreSQL.
- Integrated LLMs and OCR (Azure OpenAI, Azure Computer Vision) for email and document classification, data extraction, and structured output generation.
- Applied prompt engineering and schema validation to ensure consistent and reliable model responses.

AUTOMATED SIGNATURE & SEAL VERIFICATION SYSTEM

Python · Azure OpenAI Vision · Google Gemini · Azure Computer Vision · AWS Textract

- Designed an AI-powered document authenticity verification system for detecting signatures and seals on scanned invoices.
- Benchmarked multi-provider models (Azure OpenAI Vision, Gemini, AWS Rekognition) and established a provider-comparative evaluation framework.
- Engineered iterative prompt strategies to improve detection accuracy for faint, faded, or obscured stamps and signatures.
- Built a batch processing pipeline with retry logic and exponential backoff for folder-level invoice classification, eliminating manual verification effort.

TEST CASE GENERATION & EVALUATION API

Python · FastAPI · Hugging Face · TensorFlow/Keras · T5 · LSTM · OpenAI · LangChain

- Built a FastAPI-based system to automate generation and evaluation of BDD-style test cases from requirement documents using Gherkin inputs.
- Exposed unified REST endpoints for multi-model inference (T5, LSTM, GPT) with structured schemas, validation, and error handling for CI/CD integration.
- Implemented evaluation endpoints using BLEU, ROUGE, METEOR, and BERTScore metrics for systematic model performance comparison.
- Designed scalable request processing with seamless orchestration across multiple AI models through unified API endpoints.